

Network Neuroscience

Functional Architecture of the Human Hypothalamus: Cortical Coupling and Subregional Organization Using 7-Tesla fMRI

--Manuscript Draft--

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Abstract:	The hypothalamus is vital to the regulation of the body's metabolic state and behaviors related to survival. Despite its importance however, many questions exist regarding the intrinsic and extrinsic connections of the hypothalamus in humans, and its relationship with the cerebral cortex. Previous work on functional connectivity in the human hypothalamus have either treated it as a unitary structure or relied on methodological approaches that are limited in modeling its intrinsic functional architecture. However, the hypothalamus is a heterogeneous structure that may be composed of separate functional subunits. Here, we used resting-state data from ultra-high field 7-Tesla fMRI and a data-driven analytical approach to identify functional subregions of the human hypothalamus. We identified four functional hypothalamic subregions based on intrinsic functional connectivity with differing patterns of functional connectivity to the cerebral cortex. The anterior hypothalamic subregion showed strong connections with frontal, midline, and limbic cortical areas – many of which are canonical default mode regions. In contrast, the hypothalamus as a whole was only weakly connected to posterior sensorimotor regions. The findings support the use of ultra-high field, high-resolution imaging in providing a more incisive investigation of the human hypothalamus that respects its complex internal structure and extrinsic functional architecture.
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Additional Information:	
Question	Response

Q1: As corresponding author, I certify that my article represents original work and that there is no significant overlap with other articles that are currently under consideration or in press elsewhere.	Yes
Q2: If my article is published I agree to make data, tools and materials that are essential for reproducing the article's findings freely available. These materials are included in the Supplementary Information or are made available through a public data repository. I understand that this is an absolute requirement for Methods and Data articles.	Yes
Q3: This manuscript involves experimentation on live vertebrates and/or higher invertebrates	Yes
As corresponding author, I certify that these experiments were performed in accordance with relevant guidelines and regulations. The manuscript includes the following statement regarding approval from the relevant institutional committee, or equivalent: as follow-up to "Q3: This manuscript involves experimentation on live vertebrates and/or higher invertebrates"	The study protocol was approved by and followed the guidelines of the Institutional Review Board at Massachusetts General Hospital.
Q4: The manuscript includes data obtained from human subjects.	Yes
The manuscript contains the following statement which names and identifies the committee approving the experiments, and confirms that informed consent was obtained from all human subjects: as follow-up to "Q4: The manuscript includes data obtained from human subjects."	All recruitment, consent, and compensation procedures adhered to the IRB [of Massachusetts General Hospital]'s guidelines.
Q5: As corresponding author, I confirm that all authors have approved the manuscript in its current form.	Yes
Q6: All authors agree with publication of the article under a Creative Commons Attribution (CC BY) licence.	Yes
Q7: As corresponding author, I declare that none of the authors have any competing interests (financial, personal, or professional) that might be perceived as interfering with the objectivity of	Yes

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Q9: Were any AI tools (e.g. ChatGPT, generative AI) used to produce text or images/graphics, or to collect data?	No
Q8: Accepted manuscripts are subject to a \$2250 Article Processing Charge (APC).	I agree to pay the APC upon acceptance of the manuscript

Dear Dr. Shine,

I am writing to submit our revised manuscript “Intrinsic Functional Subregions of the Human Hypothalamus: Cortical Coupling and Subregional Organization Using 7-Tesla fMRI” [NETNEURO-25-0113] on behalf of my co-authors.

As a reminder, we used ultra-high field fMRI (7T) resting-state data to identify functional subregions within the hypothalamus and their connections to the cerebral cortex. Our approach identified four functional hypothalamic subregions. Additionally, we found that these subregions showed strong connections with a collection of prefrontal and midline regions, and weaker connectivity with a collection of more posterior sensorimotor regions, with the anterior hypothalamic subregion showing the greatest functional connectivity with the cerebral cortex overall. We have also included major revisions and new results to the manuscript based on feedback from the editor and reviewers.

As suggested by Reviewer 1, we have included the use of an older version of fmripreg as a limitation (please see our response to Reviewer 1 below). We agreed with the reviewer that this is a potential limitation and have edited the manuscript accordingly.

Once again, we would like to thank the editor and reviewers for their time and helpful feedback, which we believe have strengthened the manuscript.

Thank you,

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Reviewers' comments:

Reviewer 1:

The authors have substantially improved the manuscript and addressed several of my concerns regarding transparency, reporting, and clustering rationale.

Regarding the use of the older fMRIPrep version, the authors provide a rationale for why they believe newer versions would not materially affect the results, particularly given their distortion-matched T1wEPI approach. However, this remains a justification by argument rather than by direct empirical verification. While I appreciate the computational burden involved, the concern is not fully resolved in the absence of a benchmark analysis or stronger evidence that the relevant preprocessing steps are indeed unaffected. If the authors do not plan to address this empirically, I think this should at least be acknowledged explicitly as a limitation in the manuscript.

The exclusion of subcortical connectivity analyses is now better justified and appropriately acknowledged as a limitation. I still see this as a conceptual constraint on the biological interpretation, but the manuscript is now more transparent about that limitation.

Overall recommendation:

I support publication provided that the authors more explicitly acknowledge the residual limitation of the preprocessing choice in the manuscript, if they do not plan to address it analytically.

Response:

We thank the reviewer for their thoughtful comments on the paper and am pleased to learn we have addressed most of their major concerns.

In response to the reviewer's concern about using an older version of fmriprep we have now added it as a limitation in the revised manuscript in the Discussion section (p. 22 in the revised manuscript)

We preprocessed our data using an older version of fMRIPrep (v1.1.2). To minimize computational burden and maintain consistency with our previous reports using this dataset, we did not reprocess the data with a newer version. Although more recent releases include improvements to coregistration and fieldmap-based distortion correction, we developed and implemented a customized preprocessing pipeline tailored for high-resolution fMRI. Specifically, we acquired a structural EPI sequence with geometric distortions closely matched to those of the functional images, improving coregistration while minimizing manipulation of the functional data and obviating the need for conventional fieldmap-based distortion correction.

Intrinsic Functional Subregions of the Human Hypothalamus: Cortical Coupling and Subregional
Organization Using 7-Tesla fMRI

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Abstract

The hypothalamus is vital to the regulation of the body's metabolic state and behaviors related to survival. Despite its importance however, many questions exist regarding the intrinsic and extrinsic connections of the hypothalamus in humans, and its relationship with the cerebral cortex. Previous work on functional connectivity in the human hypothalamus have either treated it as a unitary structure or relied on methodological approaches that are limited in modeling its intrinsic functional architecture. However, the hypothalamus is a heterogeneous structure that may be composed of separate functional subunits. Here, we used resting-state data from ultra-high field 7-Tesla fMRI and a data-driven analytical approach to identify functional subregions of the human hypothalamus. We identified four functional hypothalamic subregions based on intrinsic functional connectivity with differing patterns of functional connectivity to the cerebral cortex. The anterior hypothalamic subregion showed strong connections with frontal, midline, and limbic cortical areas – many of which are canonical default mode regions. In contrast, the hypothalamus as a whole was only weakly connected to posterior sensorimotor regions. The findings support the use of ultra-high field, high-resolution imaging in providing a more incisive investigation of the human hypothalamus that respects its complex internal structure and extrinsic functional architecture.

Author Summary

Despite its importance in metabolic regulation and behavior, the functional architecture of the human hypothalamus has been surprisingly understudied. Functional connectivity studies have tended to treat the hypothalamus as a unitary structure. Yet, the hypothalamus is anatomically heterogeneous which raises questions about whether functional subunits exist within it, and their relationship with the cortex. Using resting state data from ultra-highfield (7-Tesla) functional MRI, we identified four functional hypothalamic subunits. Notably, the anterior hypothalamus, compared with other subregions, showed greater connectivity with a network of frontal and midline structures, including default mode network (DMN) regions. Our work both raises questions about how the hypothalamus and DMN regions might interact and whether the hypothalamus is better thought of as functionally heterogeneous.

Introduction

The hypothalamus is a heterogeneous structure composed of diverse nuclei that regulate various life-preserving processes such as maintaining circadian rhythms, metabolism, feeding, and thermoregulation (Burbridge et al., 2016; Lechan & Toni, 2016; Saper, 2012; Saper & Lowell, 2014). It has widespread connections with cortical areas important for sensory integration and viscerosensory processing (Barbas, 2000; Çavdar et al., 2001; Freedman et al., 2000; Kleckner et al., 2017; Öngür et al., 1998; Rempel-Clower & Barbas, 1998). These connections also position the hypothalamus to support higher-order psychological processes such as cognition, decision-making, and emotion, via links to the medial prefrontal, orbitofrontal, and anterior cingulate cortices (Burbridge et al., 2016; Goel et al., 2025; Lechan & Toni, 2016; Qu et al., 1996; Rempel-Clower & Barbas, 1998; Risold et al., 1997; Saper, 2012).

Extensive cross-species research has identified aspects of hypothalamic organization that generalize across vertebrates (Burbridge et al., 2016; King & Anthony, 1984; Löhr & Hammerschmidt, 2011; Markakis, 2002; Saper, 2012; Saper et al., 1978; Xie & Dorsky, 2017). But there are significant challenges when extrapolating this work to humans. This is especially true for corticohypothalamic connections. Unlike subcortical tissue, cortical boundaries are less clearly defined, and the non-linear expansion of the cerebral cortex across evolution complicates the identification of homologous connections—an issue that is pronounced in humans (Finlay & Uchiyama, 2015; Smaers et al., 2017). Thus, it remains unclear how much of the corticohypothalamic connections in non-human animals mirror those in humans.

Knowledge about the human hypothalamus is largely extrapolated from non-human animal work, but important anatomical differences complicate direct comparisons. Comparative anatomy has revealed both commonalities (Markakis, 2002; Saper, 2012; Xie & Dorsky, 2017) and divergences across vertebrate hypothalami (e.g., Chometton et al., 2016; King & Anthony, 1984; Panula et al., 1990). Two differences are particularly notable. First, human hypothalamic nuclei have less well-defined boundaries compared

to other mammals (Dudás, 2021; Lechan & Toni, 2016; Saper, 2012). Second, the human hypothalamus is oriented differently than in non-human animals because the human skull is proportionally shorter in the anterior-to-posterior direction. This can produce angular discrepancies of up to 45 degrees, meaning human homologues may appear anteroflexed, retroflexed, or even vertical relative to other species (Saper, 2012). Alongside variation in the location, morphology, and projections of cell populations, differences in orientation complicate the identification of homologous hypothalamic subregions in humans based on non-human animal research.

Functional magnetic resonance imaging (fMRI) offers a way to examine the functional architecture and connectivity of the human hypothalamus in vivo. Here, an important question is the extent to which fMRI can resolve functional subunits within the human hypothalamus (Lechan & Toni, 2016; Saper, 2012). Most existing functional connectivity studies have relied on 3-Tesla field-strength fMRI and have treated the hypothalamus as a single unit (e.g., Gao et al., 2020; Hinkle et al., 2013; Lukoshe et al., 2017; Moulton et al., 2014; Qiu et al., 2013; Wright et al., 2016). Yet, a couple of studies have examined hypothalamic connectivity upon dividing it into medial and lateral subunits (Hirose et al., 2016; Kullmann et al., 2014). These studies suggest that fMRI can meaningfully resolve gross hypothalamic subregions, even when using conventional 3T fMRI scanning. Extending brain imaging techniques further, Robertson et al. (2025) used high-field strength 7T fMRI, which provides a significantly greater signal to noise ratio for examining small subcortical structures in particular (e.g., periaqueductal gray, colliculi; Faull et al., 2015; Linnman et al., 2012; Satpute et al., 2013). Using a priori defined seeds, they specified six subregions of the hypothalamus which showed differential functional connectivity with prefrontal, anterior cingulate, and insular cortices.

Previous work highlights the potential of non-invasive neuroimaging methods to investigate the functional organization of the human hypothalamus. Here, we extend this line of research by adopting a data-driven approach. We combined 7T fMRI with 32-channel head coils, which offers signal-to-noise improvements nearly an order of magnitude greater than 3T methods along with increased sensitivity to microvasculature. This approach enabled data acquisition at 1.1 mm isotropic voxel resolution and has previously allowed us to identify functional subregions in small subcortical structures including the periaqueductal gray and superior colliculi (Chen et al., 2022; Fischbach et al., 2024; Kragel et al., 2021; Satpute et al., 2013; Wang et al., 2020). Rather than defining hypothalamic subregions using a priori seeds based on approximate anatomical landmarks, our aim was to determine the number and organization of functional subdivisions that can be robustly identified from the functional data itself. This approach may offer a useful complement to anatomical parcellation strategies when considering that the human hypothalamus has less well-defined internal boundaries and differs in structural organization across species (Dudás, 2021; Lechan & Toni, 2016; Saper, 2012). We then examined how functionally-defined parcels relate with cortex in particular, in part because of the difficulties in extrapolating non-human animal findings on corticohypothalamic connections to humans posed by non-linear expansion of the cerebral cortex. We present hypothalamic-subcortical connectivity in a separate manuscript.

Method

Participants

Adult, right-handed participants from the greater Boston area were recruited for the study ($N = 104$). Participants' ages ranged from 18 to 40 years ($M = 26.85$, $SD = 6.00$). Of these participants, 60 identified as male and 44 identified as female. Eleven participants identified as Hispanic or Latino, 92 identified as Non Hispanic or Latino. 59 participants identified as White or Caucasian, 13 identified as Black or African-American, 29 identified as Asian and 2 chose not to answer. For educational level: 8 had completed graduate school, 18 had completed some graduate school, 26 had completed college, 38

completed some college, 11 completed high school or a GED, and 2 had completed some high school. Ethnicity, racial, and education data were missing for 1 participant due to a technical failure in the online data collection. Of the participants, 91 had all three runs, nine participants had only two runs, and four participants had only one run. In total, the analyzed sample included 295 runs.

fMRI Acquisition

Gradient-echo echo-planar imaging BOLD-fMRI was performed on a 7 Tesla MRI scanner at the Athinoula A. Martinos Center for Biomedical Imaging at Massachusetts General Hospital (MGH), Boston, MA. The scanner was built by Magnex Scientific (Oxford, UK), with the MRI console, gradient and gradient drivers, and patient table provided by Siemens. A custom-built 32-channel radiofrequency coil head array was used for reception. Radiofrequency transmission was provided by a detunable band-pass birdcage coil. Functional images were acquired using a GRAPPA-EPI sequence (GRAPPA acceleration factor = 3, TE = 28ms, TR = 2.34s, flip angle = 75°, 123 axial slices, A > P phase encoding, partial Fourier in the phase encode direction = 7/8). Structural images were also acquired using a GRAPPA-EPI sequence (GRAPPA acceleration factor = 3, TE = 22 ms, TR = 8.52 s, flip angle = 90°, 126 axial slices, A > P phase encoding, partial Fourier in the phase encode direction = 6/8). This structural EPI image was reconstructed (via freesurfer and custom scripts) into a T1-like image, which improved anatomical-functional registration and reduced blurring of functional signals by ensuring that anatomical and functional images had similar spatial distortions (Renvall et al., 2016). We will subsequently refer to this as the T1-weighted (T1w) image for brevity. In both structural and functional images, voxels were 1.1mm isotropic (0mm gap between slices, FOV = 205 x 205 mm²), echo spacing was 0.81ms, and bandwidth was 1415 Hz per pixel.

fMRI Preprocessing

All preprocessing and analysis scripts and toolboxes can be found on the Open Science Foundation (OSF) (https://osf.io/ue7xd/overview?view_only=f072d6fe8e1b456f8754bc5c268bf308). Preprocessing of the anatomical and functional data was performed using the fmripipeline, version

1.1.2 (Esteban et al., 2019), a Nipype-based tool (Gorgolewski et al., 2011). Pipeline details can be found at <https://fmripiprep.org/en/1.1.2/workflows.html>. Each T1w volume was corrected for INU (intensity non-uniformity) using N4BiasFieldCorrection v2.1.0 (Tustison et al., 2010). Subject brain masks were computed by dilating a binary image of their skull-stripped T1w by 2 voxels to remove gaps in coverage. Spatial normalization to the 2009c ICBM 152 Nonlinear Asymmetrical template was performed through nonlinear registration with the antsRegistration tool of ANTs v2.1.0 (Avants et al., 2008), using brain-extracted versions of both T1w volume and template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1-like image using FSL fast (FSL v5.0.9; Y. Zhang et al., 2001). Functional data were slice-time corrected using 3dTshift from AFNI v16.2.07 and motion corrected using FSL mcflirt (Jenkinson et al., 2002). Mean framewise displacement was $M = .17$, $SD = .07$, interquartile range = .10. This was followed by co-registration to the corresponding T1w using boundary-based registration with 9 degrees of freedom, using FSL flirt. Motion correcting transformations, BOLD-to-T1w transformation and T1w-to-template (MNI) warp were concatenated and applied in a single step using antsApplyTransforms (ANTs v2.1.0; Cox, 1996) using Lanczos interpolation. Physiological noise regressors were extracted using the aCompCor method (Muschelli et al., 2014), taking the top five principal components from subject-specific CSF and WM masks, where the masks were generated by thresholding the WM/CSF masks derived from fast at 99% probability, constraining the CSF mask to the ventricles (using the ALVIN mask; Kempton et al., 2011), and eroding the WM mask using the binary erosion function in (SciPy v.1.6.1; Virtanen et al., 2020). Many internal operations of fmripiprep use Nilearn, principally within the BOLD-processing workflow. For all participants, the quality of brain masks, tissue segmentation, and MNI registration was visually inspected for errors using the html figures provided by the fmripiprep pipeline.

General Linear Model

In each subject, preprocessed functional BOLD data was submitted to a first-level GLM (FEAT; Woolrich et al., 2009; as implemented in `nipype 1.1.4.dev0`; Esteban et al., 2019) to remove noise attributable to motion or physiological artifacts. Nuisance regressors included 6 translation/rotation parameters, their temporal derivatives, their squares, and their squared temporal derivatives (Satterthwaite et al., 2013). Nuisance regressors also included 10 aCompCor terms, modeling the top 5 principal components of signal within whole-brain white matter and CSF masks (Muschelli et al., 2014), non-steady-state outliers (identified by `fmriprep`), and intercept, and a discrete cosine filter with a 120 second cutoff (simulating the effects of a high-pass filter within the GLM). No smoothing was performed, as all data would later be binned into ROI parcellations.

Parcellations

Hypothalamic. The probabilistic Pauli hypothalamus mask (in 2009c MNI space) was registered and resliced to native space voxel resolution (1.1 mm isotropic) using `nilearn.image.resample_img` (v0.11.1) with linear interpolation. The hypothalamic mask was then binarized using the 50% threshold of the probabilistic map, resulting in a mask with 1019 voxels. We masked out high variability voxels indicative of the third ventricle at the run level by sorting voxels based on signal variability over time and using a “kneedle” algorithm (Satopaa et al., 2011) implemented in MATLAB R2020b to obtain a threshold value. The algorithm was used to mask out voxels at a per subject and per run level. The mean number of voxels removed across the sample was $M = 43.61$ voxels with a standard deviation of $SD = 9.08$. In some cases, a voxel might be masked out due to excessive variability using the kneedle algorithm in one functional run, but not in the other ones, for a given subject’s data. In those cases, we used the available data for that voxel for analysis.

For each subject, we concatenated the run-level data across three functional runs (i.e., residuals from the GLM), and then calculated a voxel-by-voxel (i.e. 1019x1019) functional connectivity matrix using

the Pearson correlation coefficient. Because we masked out voxels with high variability on a run-by-run basis, there was the possibility that some voxels were masked out in some runs, but not others. In those cases, we used data from the available runs for analysis. We then Fisher transformed and averaged the participant-level functional connectivity matrices to estimate the group level functional connectivity matrix.

Cortical. We used the Glasser atlas (Glasser et al., 2016) for parcellation of cortical areas in volumetric space. The atlas includes 360 areas of cortical parcels (180 per hemisphere). As with the Pauli hypothalamus mask, we spatially normalized the Glasser parcels to our dataset using `nilearn.image.resample_img` (v0.11.1) with nearest neighbor interpolation for binary masks and linear interpolation for probabilistic masks (Abraham et al., 2014).

Hypothalamic Community Detection

For each subject, we concatenated the run-level data (i.e., residuals from the GLM), and then calculated a voxel-by-voxel (i.e. 1019x1019) functional connectivity matrix using the Pearson correlation coefficient. Missing values in voxels for certain time points, e.g., due to masking for a select run, were excluded from the calculation, such that voxels with data from at least one run were included in the analysis. We then Fisher transformed and averaged the participant-level functional connectivity matrices to estimate the group level functional connectivity matrix. We then performed Louvain community detection on the group level functional connectivity matrix in MATLAB R2020b, which assigns each voxel to a community. For robustness, we iterated the Louvain community detection 100 times, calculated the proportion of times each pair of voxels were assigned to the same community, and then applied the Louvain community detection algorithm again to the proportion matrix. The algorithm returned four hypothalamic communities/subregions. These communities were used as masks to extract timeseries data for each run and subject for subsequent connectivity analyses. We report an estimate of the robustness of voxel-to-community assignments by calculating the mean of proportion scores and the 25th

and 75th quartiles (see **Table 1**). The mean proportion indicates how often voxels in a given community tend to be assigned to the same community across iterations. As a check for the reliability of the number of communities identified by the Louvain algorithm, we reran the analysis for 1000 iterations without constraining the number of hypothalamic communities (see **Supplementary Figure S2**). All relevant scripts and toolboxes are available at the OSF website (https://osf.io/ue7xd/overview?view_only=f072d6fe8e1b456f8754bc5c268bf308).

Functional Connectivity Analysis

Timeseries data per ROI were concatenated across runs per subject and intercorrelated, and the Fisher z-transformed, to generate a 360 x 4 (360 cortical ROIs and 4 hypothalamic subregion ROIs) functional connectivity matrix per subject. To reduce multiple comparison concerns and simplify results, we performed a k-means cluster analysis to group together ROIs with similar connectivity profiles across hypothalamic communities (i.e. across a 360 cortical ROIs x 4 hypothalamic subregion ROIs, matrix) using `sklearn.cluster.KMeans` (v1.6.1). To identify the number of clusters that were justified by the data, we iteratively calculated the Calinski-Harabasz index using `sklearn.metrics.calinski_harabasz_score` (v1.6.1) and selected the maximum index (see **Supplemental Figure 1**). Fisher z-values for ROIs in the same k-means cluster were averaged together for each subject.

Functional Associations

To probe for functional associations of the cortical networks, we used the decoder at neurosynth.org. We uploaded the mask for each cortical network to Neurovault using the decoder tool in Neurosynth. For all cortical masks, we indicated that the map type was an ROI/mask file, the data acquisition type was fMRI-BOLD data, and that the target template was Human (MNI 152 NLin 2000c Asym). We also indicated that each mask file was resting-state data (eyes open) under the cognitive paradigm. We then examined the top ten functional terms (e.g., “conflict”, “emotional”, “faces”, “lexical decision”) and the correlation value with each network identified by neurosynth (see **Table 4**).

Results

Reliability and Robustness of the Hypothalamic Subregions

We used Louvain community detection analyses on the 1019 x 1019 functional connectivity matrix, which identified four hypothalamic subregions/communities (see **Figure 1**). We re-ran the Louvain community detection algorithm 100 times to examine the reliability of voxel assignment to these four communities, which resulted in mean probability ranging from .78 to .91 across all four communities. The quartiles indicate that the majority voxel assignments were consistently assigned to the same community. For example, voxels in the first subregion (anteroventral-tuberal hypothalamus in **Table 1**) were assigned to this community 86% of the time on average across iterations; the top quartile reached 97%, and the bottom quartile reached 82%, confirming robust voxel-to-community assignments. Additionally, to examine the reliability of the number of hypothalamic communities we conducted 1000 iterations of the Louvain community analysis which identified four hypothalamic communities in 79% of iterations (791 times), while it identified three communities in only 11% of iterations (105 times), and five communities in only 10% of iterations (104 times; see **Supplementary Figure S2**). Altogether this suggests reliability and robustness of our finding of four functional hypothalamic communities, henceforth referred to as “hypothalamic subregions”, “functional subregions”, or simply “subregions”.

Functional Subregions of the Hypothalamus

The first functional hypothalamic subregion included anteroventral portions of the anterior hypothalamus and extended to the tuberal hypothalamus (**Figure 1**, yellow). We will refer to this subregion as the Anteroventral-Tuberal subregion. The second functional subregion included mostly anterior regions of the hypothalamus (**Figure 1**, red) and thus we will refer to it as the Anterior subregion. The third functional subregion identified included areas primarily in the tuberal and posterior regions of the hypothalamus. In the superior-to-inferior plane, this subregion was located in the middle of the hypothalamus (**Figure 1**, green). We will refer to this subregion as the Middle Tuberal-Posterior subregion.

Finally, the fourth functional subregion included the superior parts of the hypothalamus and we thus call it the Superior Hypothalamic subregion(**Figure 1**, blue).

Cortical Connectivity with Hypothalamic Subclusters

The Calinski-Harabasz index showed a max value at a two-network solution (see **Supplementary Figure S1**) and the index remained high at a four-network solution. Thus, we report the results of both solutions for the sake of completeness.

Two-Network Solution

The first cortical network (orange in **Figure 3a**) contained primarily anterior and midline regions. Regions in Cortical Network 1 included the OFC, ventromedial and dorsomedial PFC, ventrolateral PFC, cingulate cortex, and precuneus. They also included an anterior portion of the superior temporal gyrus, the insula, and both medial and lateral portions of the superior frontal gyrus (SFG) and middle frontal gyrus (MFG). The second network (teal in **Figure 3a**) included mostly lateral and posterior brain regions, though it also included a more ventral part of the dorsomedial PFC. Other regions included in Cortical Network 2 were the dorsolateral PFC and broad swathes of sensorimotor areas, such as the pre- and postcentral gyri, temporal lobe, parietal lobe, and the occipital lobe.

We probed for differences between the cortical networks in their functional connectivity with the different hypothalamic subregions using a 2 (cortical networks) x 4 (hypothalamic subregions) repeated measures analysis of variance (ANOVA). Mean correlation values, standard deviations, and 95% CI for the relationship between each cortical network and hypothalamic subregion are reported in **Table 2**. We found a main effect of cortical network, $F(1, 103) = 175.57, p < .001$, showing that Cortical Network 1 had greater connectivity with the hypothalamic subregions ($M = .12, SD = .10$) than Cortical Network 2 ($M = .03, SD = .06$). We also found a main effect of hypothalamic subregion, $F(2.12, 217.99) = 29.60, p < .001$, suggesting that there were differences between the subregions in their connections with the cortical networks.

These main effects were qualified by a significant two-way interaction, $F(2.24, 230.80) = 33.28, p < .001$. Cortical Network 1 was significantly related to all hypothalamic subregions (see **Table 2** and **Figure 3b**), whereas Cortical Network 2 had significant connectivity with all hypothalamic subregions except the anteroventral-tuberal hypothalamic subregion (see **Table 2** and **Figure 3b**). We also conducted post-hoc Least Significant Difference (LSD) tests comparing the connectivity of each hypothalamic subregion with each cortical network. The hypothalamic subregion most strongly connected with Cortical Network 1 ($M_{diff} \geq .04, p's < .001$) and Cortical Network 2 ($M_{diff} \geq .06, p's < .001$) was the anterior hypothalamic subregion in pairwise comparisons of the functional connectivity between each subregion and each cortical network (see **Table 2** and **Figure 3b**). This was followed by the middle tuberal-posterior hypothalamus, which showed stronger connectivity with Cortical Network 1 compared to the anteroventral-tuberal ($M_{diff} = .04, p = .003$) and superior subregions ($M_{diff} = .03, p < .001$). However, the middle tuberal-posterior was only marginally greater than the anteroventral-tuberal hypothalamus ($M_{diff} = .02, p = .05$) in its connectivity with Cortical Network 2, and equivalent in connectivity as superior subregions ($M_{diff} = .01, p = .22$).

Four-Network Solution

With the four-network solution, it is notable that the four networks appear to reflect subnetworks within the cortical networks of the two-network solution (see **Figure 4a**). The first two cortical networks appear to be nested within Cortical Network 1 of the two-network solution. Thus, we have named them Cortical Network 1a and Cortical Network 1b. Cortical Network 1a, (orange in **Figure 4a**) included somatosensory association areas, the pregenual anterior cingulate cortex, and both medial and lateral parts of the prefrontal cortex. Cortical Network 1a also included medial posterior cingulate. To probe the functional relevance of the cortical network analyses, we then used the decoder tool on neurosynth.org to identify functional terms associated with regions included in each cortical network. For Cortical Network 1a, the decoder results that showed this network included areas associated with terms relating to conflict, affect, olfaction, and gustation (see **Table 4**).

Cortical Network 1b (yellow in **Figure 4a**) was the most constrained of the four networks. Cortical Network 1b included anterior brain areas and was confined to medial and lateral areas in the frontal lobe. Specifically, Cortical Network 1b network included the superior lateral and medial frontal cortex, with some areas in the basal frontal cortex. The neurosynth.org decoder analysis revealed that Cortical Network 1b included cortical areas associated with terms related to memory, particularly episodic memory and memory retrieval (see **Table 4**).

The third and fourth cortical networks were similarly nested within Cortical Network 2 of the two-network solution and are likewise named Cortical Network 2a and Cortical Network 2b. Cortical Network 2a (teal in **Figure 4a**) was highly dispersed and with a few exceptions primarily included lateral portions of the inferior temporal, superior temporal areas, and some superior frontal areas. However, Cortical Network 2a also included some medial portions of the occipital cortex and a large portion of the temporal cortex inferior to the frontal lobe. The neurosynth.org decoder results showed that Cortical Network 2a included regions associated with terms related to visual processing and particularly processing of faces (see **Table 4**).

Finally Cortical Network 2b (blue in **Figure 4a**) included primarily posterior areas, including many visual areas in occipital cortex and auditory areas in temporal cortex. Cortical Network 2b also extended anteriorly to include the postcentral gyrus, parts of lateral frontal areas. Finally, Cortical Network 2b included midline areas such as the cuneus and precuneus and the subgenual anterior cingulate cortex in the right hemisphere. The results of the neurosynth.org decoder analysis revealed that the regions included in Cortical Network 2b are regions that are associated with terms related to auditory linguistic processing and theory of mind (see **Table 4**).

As with the two-network solution, we next probed for differences in the four cortical networks' connectivity with the hypothalamic subregions in a 4 (cortical networks) x 4 (hypothalamic subregions) repeated measures ANOVA. Mean correlation values, standard deviations, and 95% CI for the relationship

between each cortical network and hypothalamic subregion are reported in **Table 3**. We found a main effect of cortical network, $F(1.11, 113.99) = 152.03, p < .001$, and hypothalamic subregion, $F(2.12, 217.80) = 29.56, p < .001$. These results suggest differences in the connectivity between both the cortical networks and hypothalamic subregions. These main effects were qualified by a significant two-way interaction, $F(3.19, 329.02) = 27.80, p < .001$.

Cortical Networks 1a, 1b, and 2b all showed significant connectivity with all subregions of the hypothalamus ($p's < .001$), whereas Cortical Network 2a only showed significant connectivity only with the anterior hypothalamus ($p < .001$; see **Table 3** and **Figure 4b**). This difference is what likely drove the interaction between the cortical networks and hypothalamic subregions. Post-hoc LSD revealed that Cortical Network 1a showed the strongest connectivity with all hypothalamic subregions compared to the other cortical networks ($M_{diff} \geq .05, p's < .001$). Cortical Network 1b had the second strongest connectivity, having greater connectivity with all hypothalamic subregions than both Cortical Networks 2a ($M_{diff} \geq .06, p's < .001$) and 2b ($M_{diff} \geq .04, p's < .001$). Meanwhile, Cortical Network 2a had the weakest functional connectivity overall (see **Table 3** and **Figure 4b**).

As with the two-network solution, the anterior hypothalamus also showed the strongest connectivity with all four cortical networks compared to the connectivity of other hypothalamic subregions ($M_{diff} \geq .05, p's < .001$), followed by middle tuberal-posterior hypothalamus (see **Table 3** and **Figure 4b**). More specifically, the middle tuberal-posterior hypothalamus showed greater connectivity with Cortical Networks 1a and 1b than the anteroventral-tuberal ($M_{diff} \geq .03, p's < .001$) and superior subregions ($M_{diff} \geq .03, p's < .001$). Meanwhile, the middle tuberal-posterior hypothalamus showed greater functional connectivity with Cortical Network 2b than the anteroventral-tuberal hypothalamus ($M_{diff} = .03, p = .02$), but was equivalent in its strength of connectivity with the superior hypothalamus ($M_{diff} = .01, p = .15$).

Discussion

The present study suggests that ultra-high field, 7-Tesla resting-state fMRI combined with a data-driven analytical approach can robustly resolve the human hypothalamus into four functional subregions with distinct patterns of cortical functional connectivity. These subregions are best viewed as a data-driven parcellation given the boundaries of the technique. Importantly, the hypothalamus does not conform to a single organizational scheme: cytoarchitecture, chemoarchitecture, connectivity, and functional specialization each emphasize partially distinct forms of organization. Within these constraints, the present findings provide insight into the level of hypothalamic organization that can be reliably characterized using ultra high field strength magnets. The subregions can be broadly divided into anteroventral-tuberal, anterior, middle tuberal-posterior, and superior hypothalamic subregions.

All four subregions showed overall greater functional connectivity with anterior medial PFC, cingulate cortex, superior frontal gyrus, and lateral orbitofrontal cortex (labeled as Cortical Network 1 and which fractionates into Cortical Networks 1a and 1b; see **Figures 3a** and **4a**). These findings parallel known cortico-hypothalamic structural connections observed in non-human animals (Öngür et al., 1998; Rempel-Clover & Barbas, 1998; Risold et al., 1997). Critically, a significant interaction indicated that the relationship between hypothalamic subregion to cortical connectivity depended on the hypothalamic subregion and cortical network, highlighting both the particularly strong relationship of the anterior subregion, and a generally weak relationship of the anteroventral-tuberal subregion, with these cortical networks. (see **Figures 3b** and **4b**).

Cortical networks that were defined by their relationship with hypothalamic subregions were associated with different term-based decoding maps. Cortical Network 1a was associated with affective, social, gustatory, and olfactory processing, while Cortical Network 1b was associated with episodic and autobiographical memory (see **Table 4**). All four subregions also showed relatively lower functional connectivity with a second collection (Cortical Network 2; Cortical Networks 2a and 2b) of largely posterior

cortical areas and dorsolateral prefrontal cortex, with Cortical Network 2a associated with visual processing and face/object recognition, and Cortical Network 2b associated with auditory and lexical processing as well as theory of mind.

Intrinsic Connectivity Captures Functional Subregions of the Human Hypothalamus

The hypothalamus is a heterogeneous structure composed of distinct nuclei with specialized functions (Lechan & Toni, 2016; Saper, 2012; Saper & Lowell, 2014) and our results indicate that this heterogeneity is reflected in the intrinsic and extrinsic functional connections of the hypothalamus. Most fMRI studies have used 3T (or less) fMRI and treated the hypothalamus as a unitary structure (e.g., Gao et al., 2020; Hinkle et al., 2013; Lukoshe et al., 2017; Moulton et al., 2014; Qiu et al., 2013; Wright et al., 2016), while a few used medial-lateral (Hirose et al., 2016; Kullmann et al., 2014) or anterior-posterior subdivisions (e.g., Schulte et al., 2017; Smeets et al., 2005). In contrast, our data-driven approach identifies functional subregions based on intrinsic connectivity. Although we did not find a strong lateral-to-medial division, our functional subregions appear to align with manually segmented structural subdivisions of the hypothalamus (Billot et al., 2020; Makris et al., 2013; Rodrigues et al., 2024). The four-subregion parcellation may serve as seeds, perhaps in conjunction with anatomical models, to guide future work on the functional architecture of the human hypothalamus, including studies using more readily available 3T methods.

Functional Connectivity between Hypothalamic Subregions and the Cerebral Cortex

The hypothalamus as a whole, and the anterior hypothalamic subregion in particular, were strongly connected to a collection of prefrontal and midline cortical areas. These results are consistent with known anatomical corticohypothalamic connections in non-human animals in which prefrontal and insular areas send direct projections to preoptic and anterior hypothalamic nuclei in rodents (Risold et al., 1997; Saper, 2012). Our findings also converge with seed-based, resting-state human neuroimaging studies that similarly found functional connections between the hypothalamus and prefrontal, insular,

and midline regions (Hirose et al., 2016; Kullmann et al., 2014; Robertson et al., 2025). A possibility is that the pattern of functional corticohypothalamic connectivity we and others have observed might be partly undergirded by homologous anatomical connections in humans.

The keyword analysis showed that the network with the strongest functional connectivity to the hypothalamus, Cortical Network 1a, was correlated with affective, olfactory, and gustatory processing. This network included the orbitofrontal cortex, anterior cingulate cortex, and insula which are areas implicated in feeding behavior, affect, and viscerosensory processing (Caseras et al., 2013; Rolls et al., 2003; Zaki et al., 2012). Cortical Network 1b, the second most strongly connected, was correlated with episodic memory and memory retrieval and included medial and lateral prefrontal cortices and parahippocampal cortex (Euston et al., 2012; Preston & Eichenbaum, 2013). These findings accord with the hypothalamus's role in metabolic regulation (Lechan & Toni, 2016; Saper, 2012) and memory consolidation and updating (Burdakov & Peleg-Raibstein, 2020, 2025; Silva et al., 2016). Although speculative, one possibility is that the hypothalamus supports the encoding or deployment of memory in service of regulating the body's metabolic state. It may draw on stored memories to anticipate future needs given current sensory input (see Schulkin, 2003; Sterling, 2004, 2012). Such an interpretation would be consistent with predictive processing models of affect (Barrett, 2017; Hesp et al., 2019; Lee et al., 2021; Seth, 2013), in which the brain proactively regulates the body's metabolic state using information stored in memory.

Hypothalamic Connectivity with the Default Mode Network and Allostatic Network

Several prefrontal cortical areas with strong functional connectivity with the hypothalamus are part of the default mode network, a set of regions associated with episodic memory, language, social cognition, and emotion representation (Barrett & Satpute, 2013; Biswal & Uddin, 2025; Buckner & Carroll, 2007; Raichle, 2015). The default mode network is not monolithic but comprises several networks and states (Andrews-Hanna et al., 2010; Buckner & DiNicola, 2019; Ciric et al., 2017; Ranganath & Ritchey,

2012; Satpute & Lee, 2026). Recent cytoarchitectonic findings suggest that more anterior nodes—including anteromedial and ventrolateral PFC—are more isolated from exteroceptive sensory input relative to more posterior nodes (Paquola et al., 2025). We found that these anterior prefrontal regions exhibited stronger connectivity with the hypothalamus, suggesting preferential involvement in viscerosensory and visceromotor processes (Kleckner et al., 2017). This anterior subset also clustered with regions known for bodily regulation—anterior and posterior cingulate cortex and insula—reinforcing the idea that functional differentiation within the default mode network may reflect differences in connections to external versus visceral sensory systems (Satpute & Lee, 2026).

These findings also align with the formulation of an allostatic network for predictive regulation of the body's internal state (Girn et al., 2024; Kleckner et al., 2017; Zhang et al., 2025; also see a “revised limbic network” model, Catani et al., 2013). By this account, functional coupling between the hypothalamus and this network may reflect the dynamic interaction between cortical predictions and prediction errors arising from the hypothalamus and brainstem nuclei (Barrett & Simmons, 2015; Chanes & Barrett, 2016; Kleckner et al., 2017).

Finally, the anterior hypothalamic subregion in our results likely contains the preoptic and anterior hypothalamic nuclei, which play a central role in thermoregulation via metabolic, autonomic, and behavioral strategies (Morrison & Nakamura, 2011, 2019; Mota-Rojas et al., 2021; Siemens & Kamm, 2018). Thermoregulation represents one of the most basic and continuous allostatic challenges an organism faces (Angilletta et al., 2010; Huey & Stevenson, 1979), and the relatively chronic need to monitor body temperature may help explain the stronger functional connectivity of the anterior hypothalamus with prefrontal, cingulate, and insular portions of the allostatic network.

Limitations and Future Directions

Our work focused on a group-level, resting-state approach to identify modal patterns of intrinsic and extrinsic hypothalamic connectivity. Future work might build on these results by examining

individual differences in functional subregion number or cortical network connectivity (Geerligs et al., 2015; Mueller et al., 2013; Vanderwal et al., 2017) and state-dependent variability (Ciric et al., 2017; Cole et al., 2021; Elton & Gao, 2015). It may also be of interest to examine whether hypothalamic-cortical connectivity correlates with individual difference measures such as anxiety or stress.

Additionally, although we treated voxels as belonging exclusively to a single subregion, voxels may belong to more than one functional subregion, and this membership may vary across states or task demands. We only relied on the Pauli hypothalamus masks to estimate the outer boundaries of the hypothalamus. However, we found that clustering was located medially within the hypothalamus, making it unlikely that our findings were due to boundary effects. We preprocessed our data using an older version of fMRIPrep (v1.1.2). To minimize computational burden and maintain consistency with our previous reports using this dataset, we did not reprocess the data with a newer version. Although more recent releases include improvements to coregistration and fieldmap-based distortion correction, we developed and implemented a customized preprocessing pipeline tailored for high-resolution fMRI. Specifically, we acquired a structural EPI sequence with geometric distortions closely matched to those of the functional images, improving coregistration while minimizing manipulation of the functional data and obviating the need for conventional fieldmap-based distortion correction. Finally, our analyses focused specifically on corticohypothalamic functional connections; future work should examine the functional relationship between these subregions and subcortical structures, given the hypothalamus's widespread subcortical connectivity. We relied on the Pauli mask to estimate the outer boundary of the hypothalamus.

Conclusion

We used high-resolution, high-field strength brain imaging to provide the first data-driven functional parcellation of the human hypothalamus using resting-state data. The results justified four functional divisions, which showed differential functional connectivity with two large cortical networks,

each further fractionable into two smaller networks. The anterior showed the greatest functional connectivity with prefrontal and limbic/paralimbic cortical areas—including nodes of the default mode network and an allostatic network. These functional parcellations and their relations to cortical network architectures may guide future work on how the hypothalamus and its functional-anatomic complexity support the integration of mind, brain, and body.

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Tables

Hypothalamic Subregion	<i>N</i> voxels	Mean Probability [25 th P, 75 th P]
Anteroventral-Tuberal Hypothalamus	295	.86 [.82,.97]
Anterior Hypothalamus	234	.91 [.88, 1.00]
Middle Tuberal-Posterior Hypothalamus	291	.74 [.63, .88]
Superior Hypothalamus	199	.88 [.74, .99]

Table 1. Robustness of voxel assignment to hypothalamic subregion. **Table 1** shows the mean probability of a given pair of voxels being assigned to the same (vs. a different) subregion across 100 iterations of Louvain Community Detection analysis. Also shown are the 25th and 75th percentiles of those probabilities in brackets and the number of voxels within each hypothalamic subregion. P = Percentile.

	Mean <i>r</i> -value	SD	95% CI
Cortical Network 1			
Anteroventral-Tuberal	.08	.14	[.06, .11]
Anterior	.18	.11	[.16, .20]
Middle Tuberal-Posterior	.12	.10	[.10, .14]
Superior	.09	.11	[.07, .11]
Cortical Network 2			
Anteroventral-Tuberal	.01	.10	[-.01, .03]
Anterior	.06	.07	[.05, .08]
Middle Tuberal-Posterior	.03	.07	[.01, .04]
Superior	.02	.08	[.003, .03]

Table 2. Table displaying mean *r*-values, standard deviations, and 95% confidence intervals for each cortical network by hypothalamic subregion in the two-network solution.

	Mean <i>r</i> -value	SD	95% CI
Cortical Network 1a			
Anteroventral-Tuberal	.12	.16	[.09, .15]
Anterior	.21	.13	[.19, .24]
Middle Tuberal-Posterior	.15	.12	[.12, .17]
Superior	.12	.13	[.09, .15]
Cortical Network 1b			
Anteroventral-Tuberal	.06	.13	[.03, .08]
Anterior	.15	.09	[.13, .16]
Middle Tuberal-Posterior	.10	.08	[.08, .12]
Superior	.07	.09	[.05, .09]
Cortical Network 2a			
Anteroventral-Tuberal	-.01	.10	[-.03, .02]
Anterior	.04	.08	[.02, .05]
Middle Tuberal-Posterior	.003	.07	[-.01, .02]
Superior	-.002	.08	[-.02, .01]
Cortical Network 2b			
Anteroventral-Tuberal	.02	.11	[-.001, .04]
Anterior	.09	.07	[.07, .10]
Middle Tuberal-Posterior	.05	.07	[.03, .06]
Superior	.03	.08	[.02, .05]

Table 3. Table displaying mean *r*-values, standard deviations, and 95% confidence intervals for each cortical network by hypothalamic subregion in the four-network solution.

Cortical Network 1a		Cortical Network 1b		Cortical Network 2a		Cortical Network 2b	
Term	Corr	Term	Corr	Term	Corr	Term	Corr
conflict	.15	episodic memory	.08	objects	.20	speech	.13
olfactory	.12	episodic	.07	visual	.18	theory of mind	.13
social	.08	memory	.07	face recognition	.17	voice	.13
food	.08	executive	.07	face	.17	listening	.12
disgust	.08	autobiographical	.06	object	.17	mind	.12
emotional	.07	planning	.06	orthographic	.15	speaker	.12
moral	.07	memory retrieval	.06	recognition	.15	listen	.12
taste	.07	remembering	.06	faces	.14	sentences	.11

Table 4. Top ten functional terms and their correlations associated with each cortical network obtained from Neurosynth.

Figures

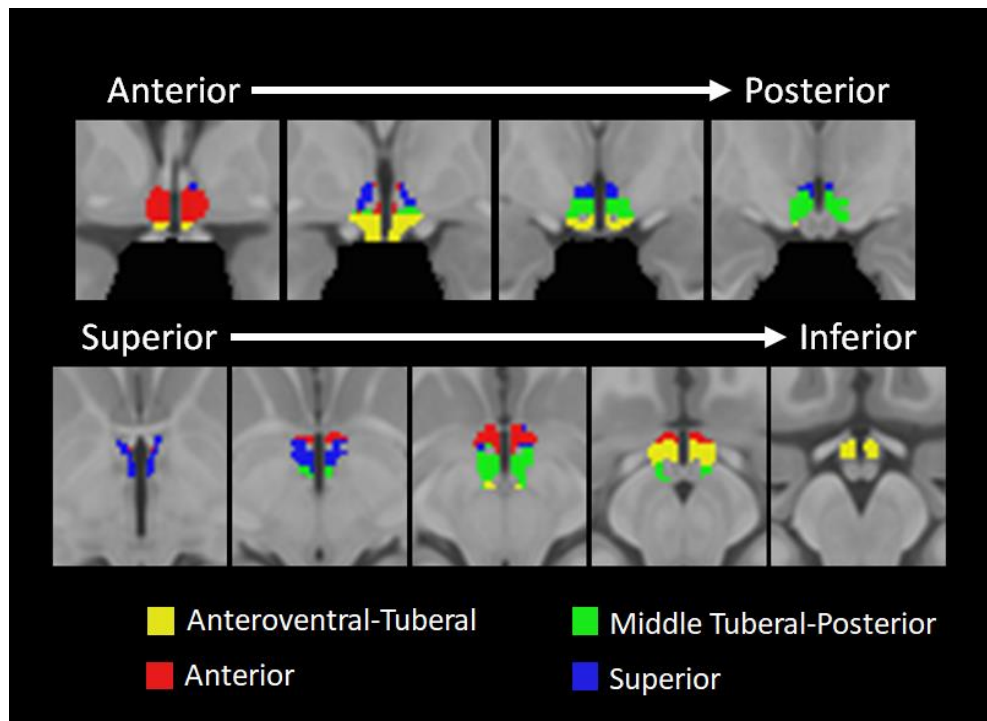


Figure 1. The four hypothalamic subregions identified using community detection analysis and two cortical networks identified with k-means clustering analysis. **Figure 1**, top row shows the coronal view of the hypothalamic subregions and moves from anterior to posterior (left to right). **Figure 1**, bottom row shows the axial view of the hypothalamic subregions moving from superior to inferior (left to right). In the figure, the anteroventral-tuberal subregion is depicted in yellow, the anterior subregion is depicted in red, the middle tuberal-posterior subregion is depicted in green, and the superior subregion is depicted in blue.

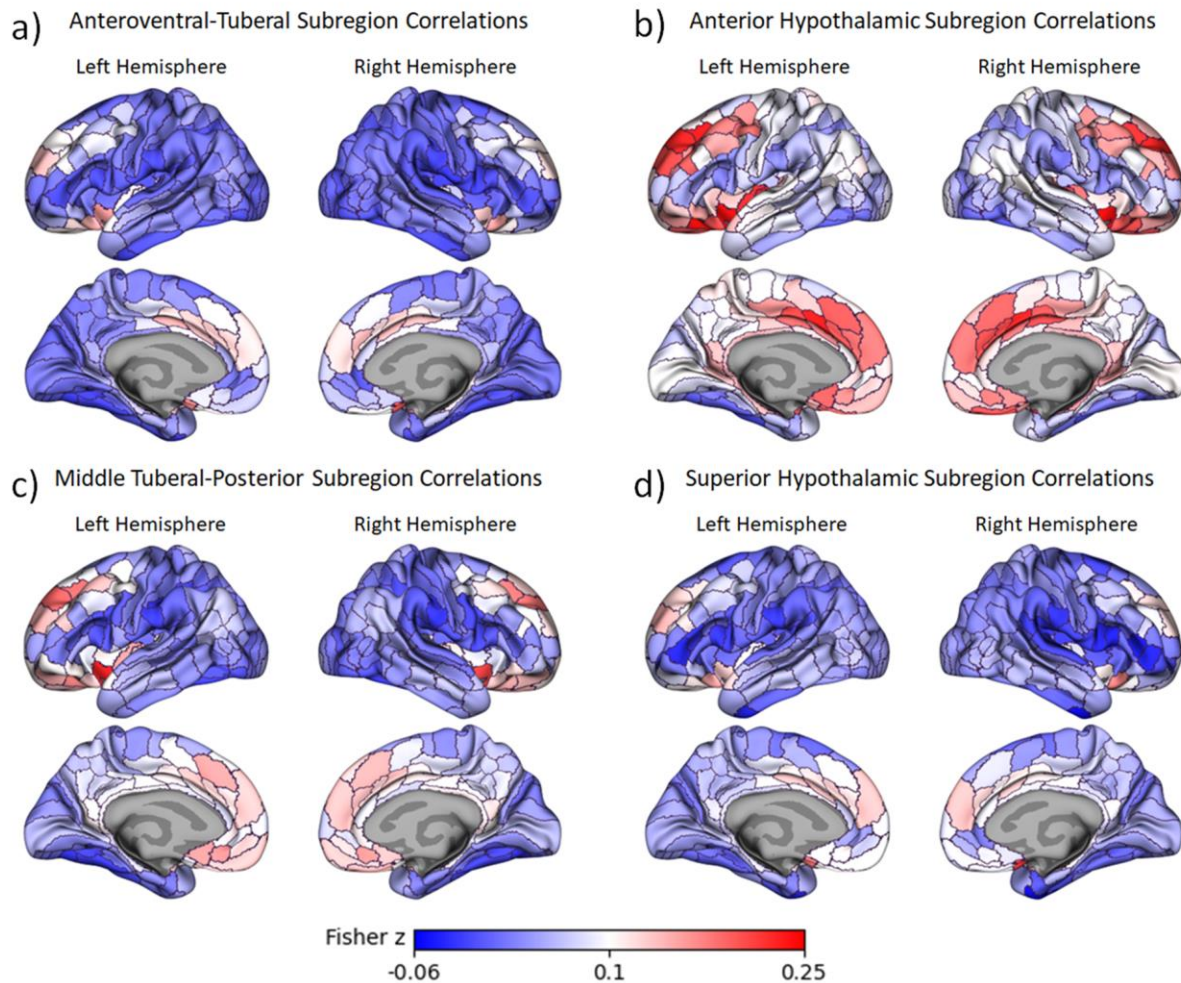


Figure 2. Correlations in Fisher's z between each hypothalamic subregion and each cortical parcel. Correlations for the anteroventral-tuberal subregion are shown in **2a**, correlations for the anterior subregion are shown in **2b**, correlations for the middle tuberal-posterior subregion are in **2c**, and correlations for the superior subregion are depicted in **2d**. Red depicts stronger positive correlations while blue depicts stronger negative correlations. In each figure, the left hemisphere is shown on the left and the right hemisphere is shown on the right. The lateral views are on the top row, while the medial views are on the bottom row.

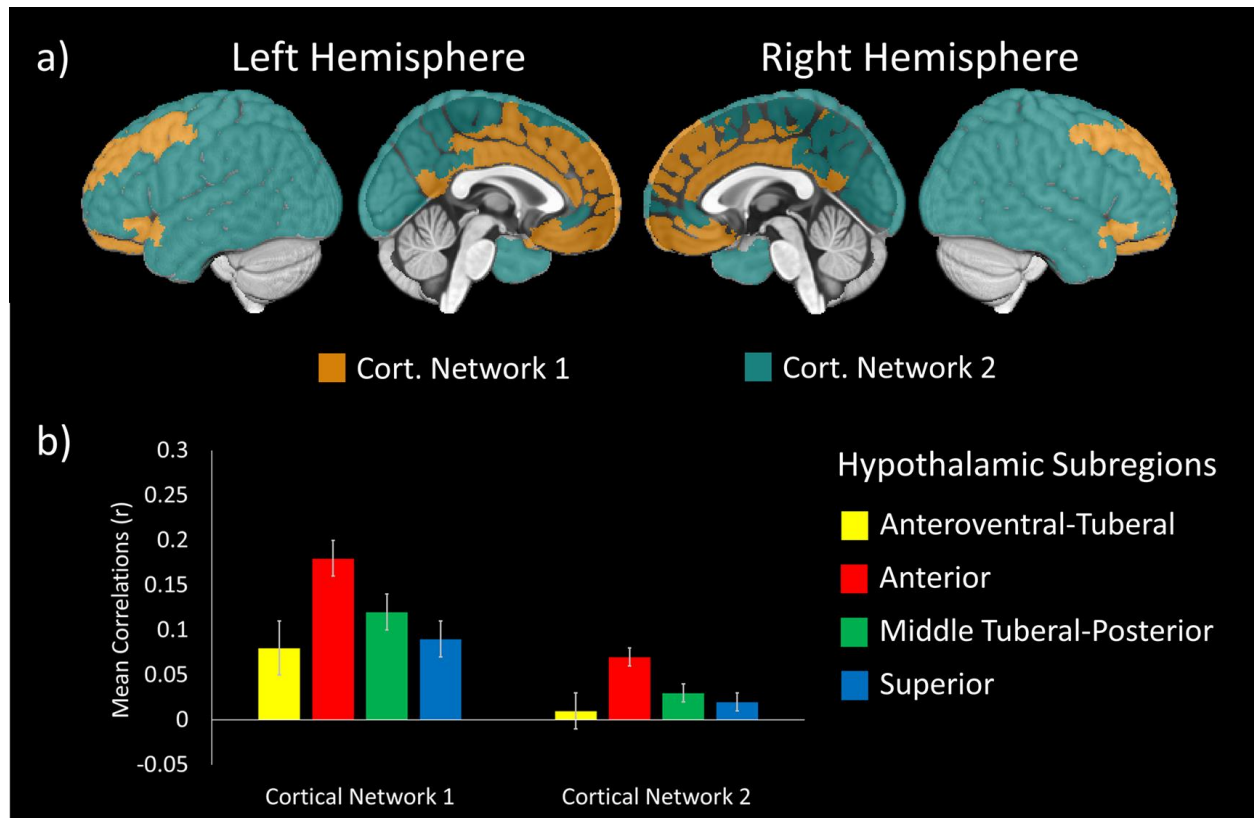


Figure 3 shows the two-network functional connectivity map and correlations between the cortical parcels and the hypothalamus. **Figure 3a** shows the cortical networks with Cortical Network 1 is in orange and Cortical Network 2 is in teal. **Figure 3b** shows the mean correlations between each cortical network and each hypothalamic subregion in the two-network solution. Each cortical network is plotted on the x-axis, while each hypothalamic subregion is depicted in different colors. The mean correlations are on the y-axis. Error bars reflect 95% confidence intervals. Overall, the hypothalamic subregions showed greater connectivity with frontal and midline regions in Cortical Network 1 compared to the more posterior and lateral sensorimotor areas in Cortical Network 2. Moreover, the anterior hypothalamic subregion (Red) showed the greatest connectivity with both cortical networks compared to all other hypothalamic subregions

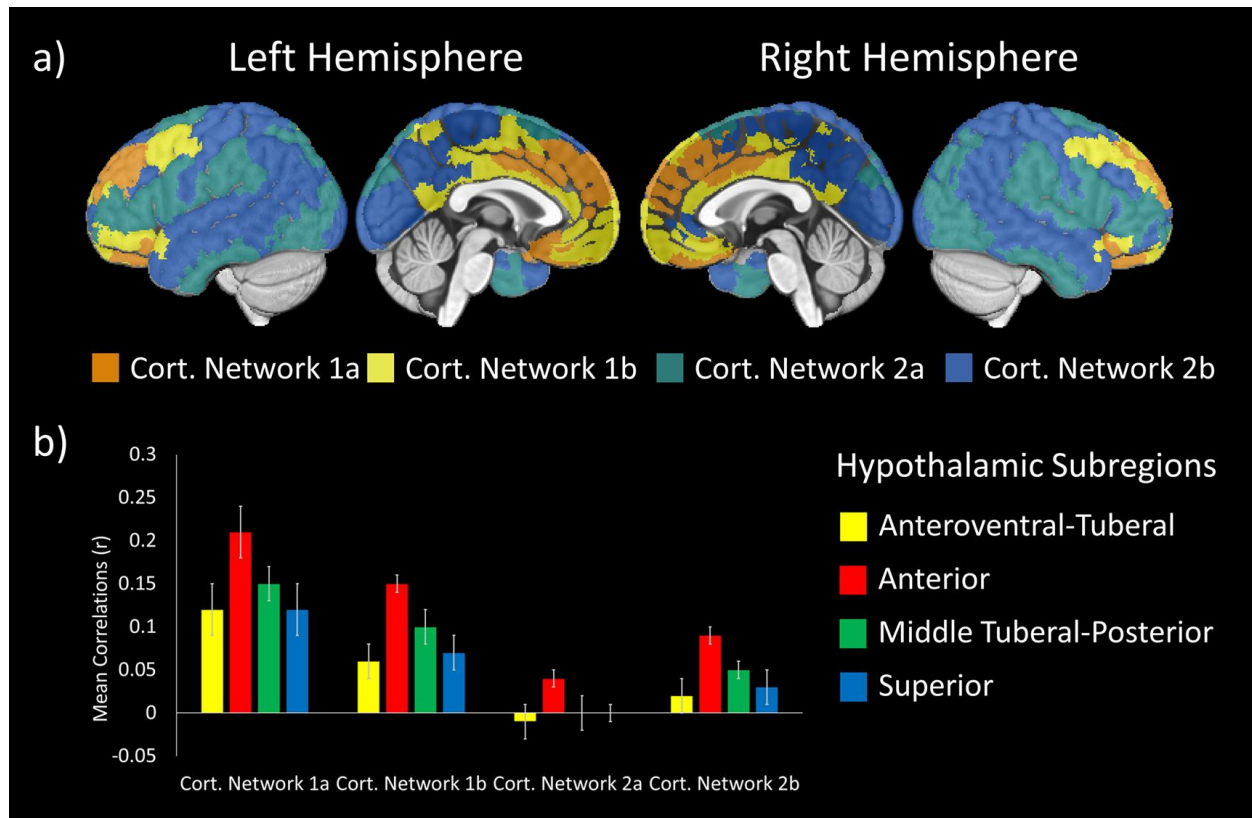


Figure 4 shows the four network functional connectivity map and correlations between the cortical parcels and the hypothalamus. **Figure 4a** shows the cortical networks with Cortical Network 1a in orange, Cortical Network 1b in yellow, Cortical Network 2a in teal, and Cortical Network 2b in blue. **Figure 4b** shows the mean correlations between each cortical network and each hypothalamic subregion in the four-network solution. Each cortical network is plotted on the x-axis, while each hypothalamic subregion is depicted in different colors. The mean correlations are on the y-axis. Error bars reflect 95% confidence intervals. Overall, the hypothalamic subregions showed greater connectivity with frontal and midline regions in Cortical Networks 1a and 1b compared to the more posterior and lateral sensorimotor areas in Cortical Networks 2a and 2b. Moreover, the anterior hypothalamic subregion (Red) showed the greatest connectivity with both cortical networks compared to all other hypothalamic subregions.

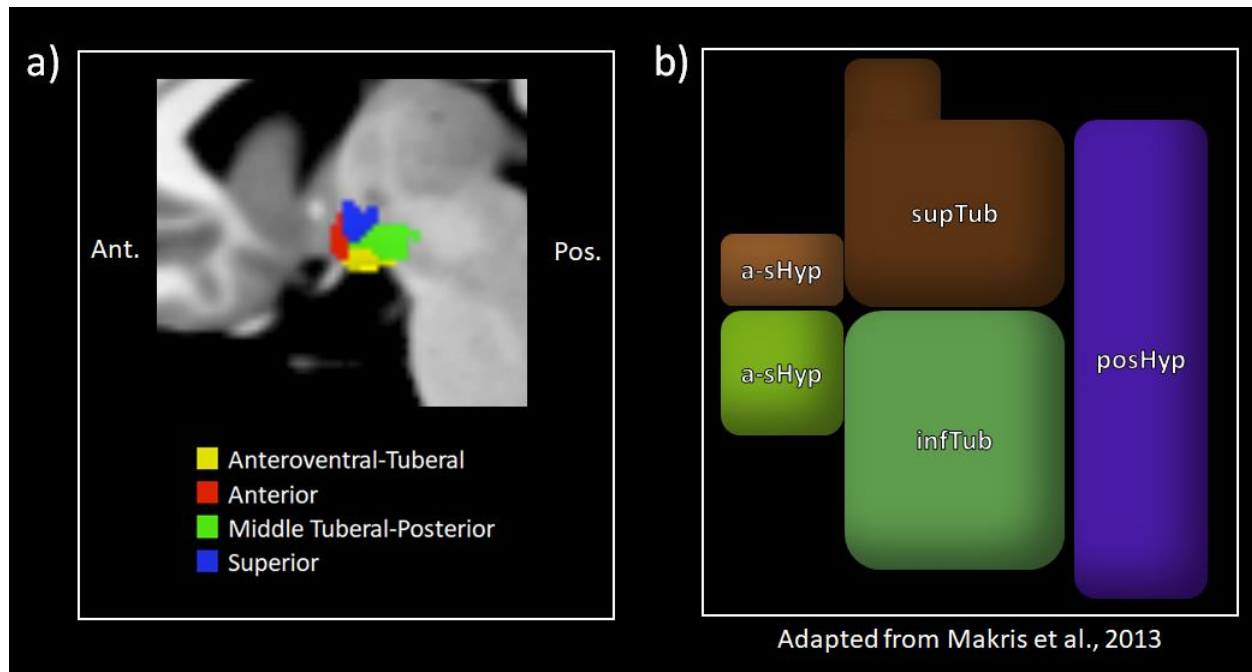
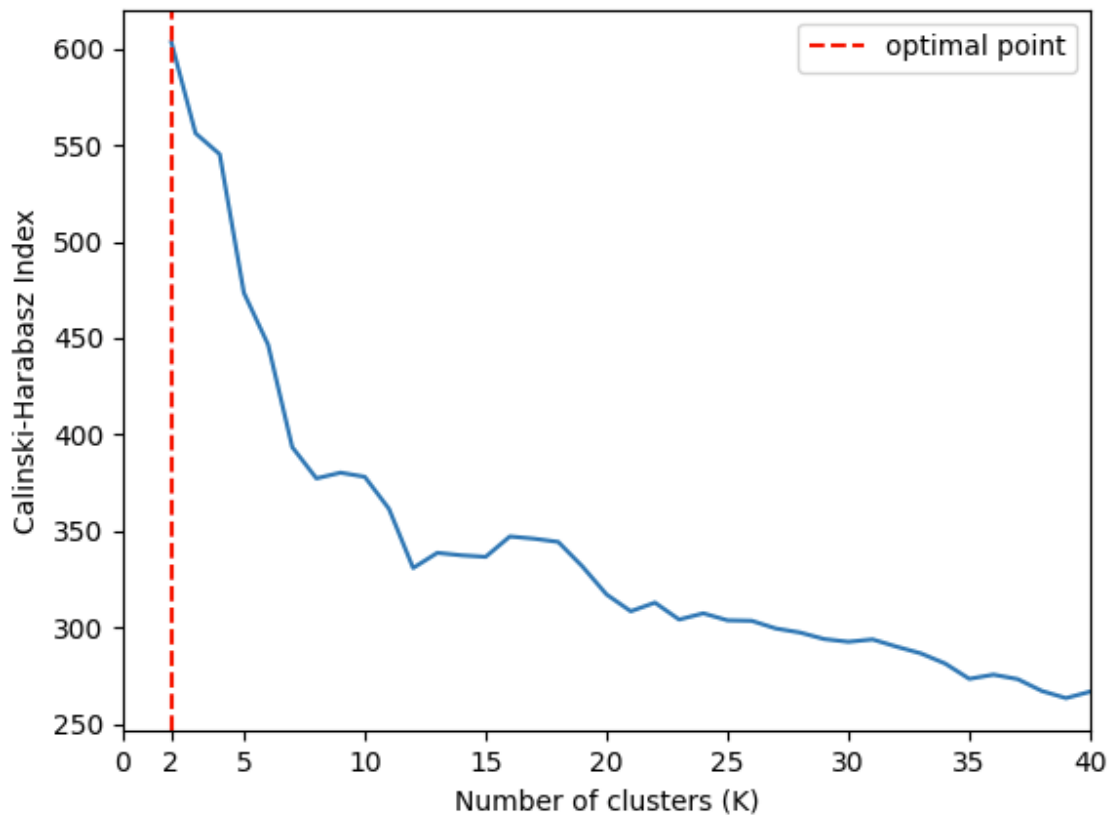
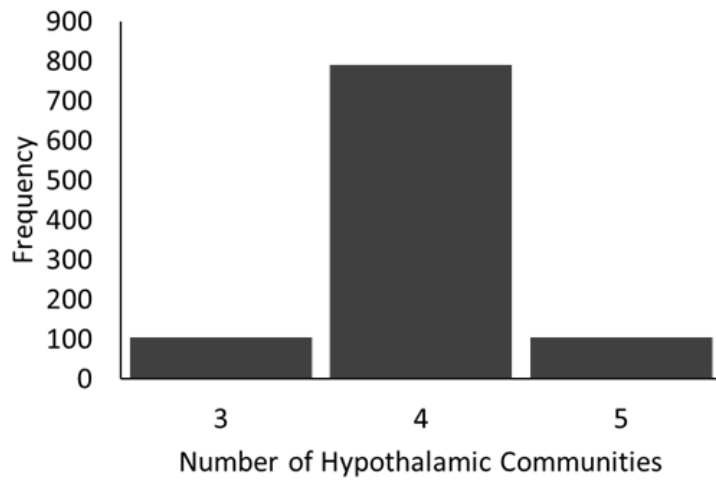


Figure 5. Comparison of functionally derived subregions in this study and anatomical parcels made by Makris et al. 2013. **4a** depicts the functional subregions identified in the present study in the medial view from the left hemisphere. and **4b** depicts Makris et al.'s (2013; Figure 2) parcellation of the hypothalamus based on anatomical landmarks, adapted from their paper. Note that the third ventricle is visible in the figure from Makris et al. (2013) in tan in. Ant. = Anterior, Pos. = Posterior, a-sHyp = anterior-superior hypothalamus, a-iHyp = anterior-inferior hypothalamus, supTub = superior tuberal hypothalamus, infTub = inferior tuberal hypothalamus, posHyp = posterior hypothalamus

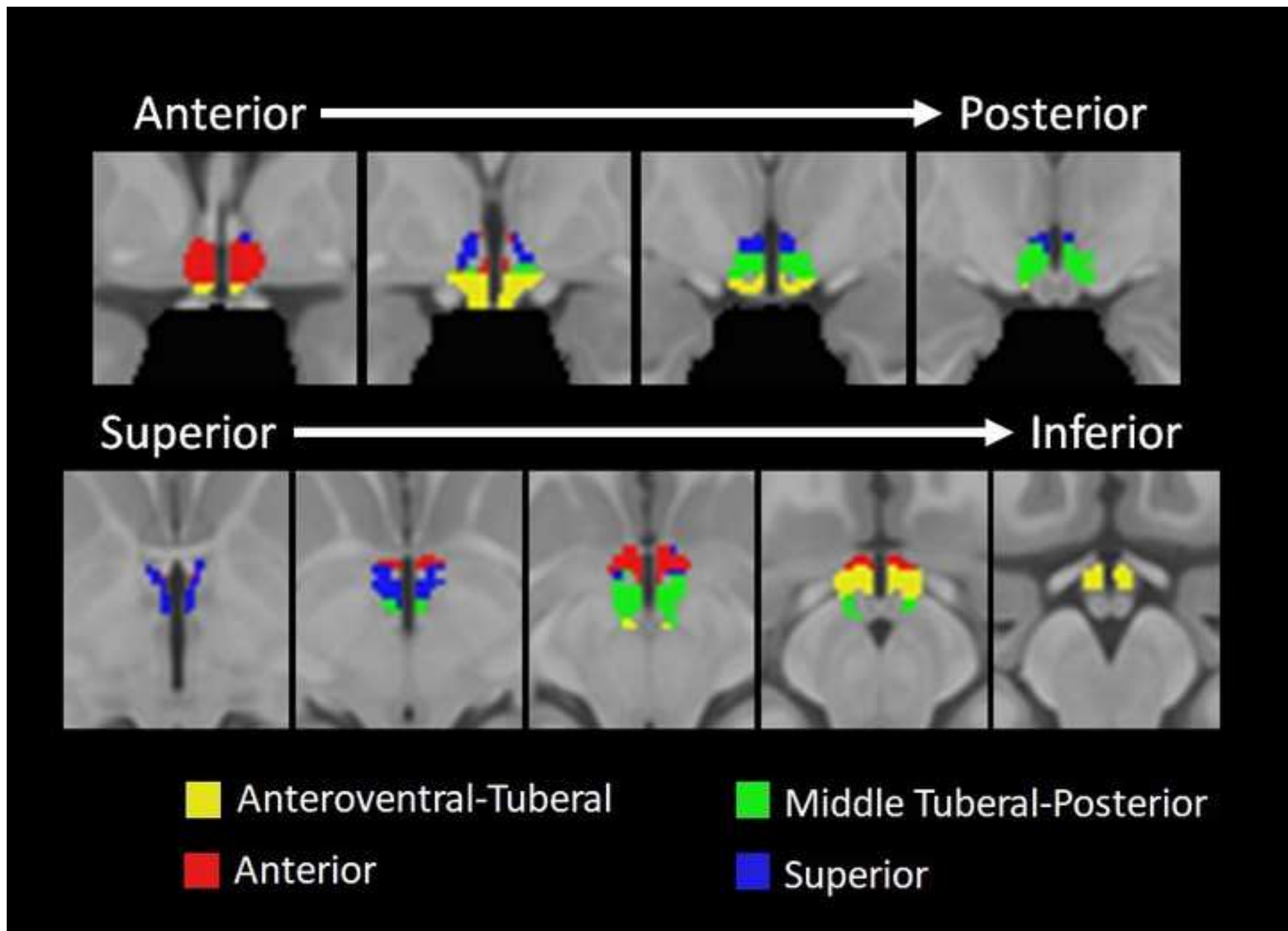
Supplementary Figure

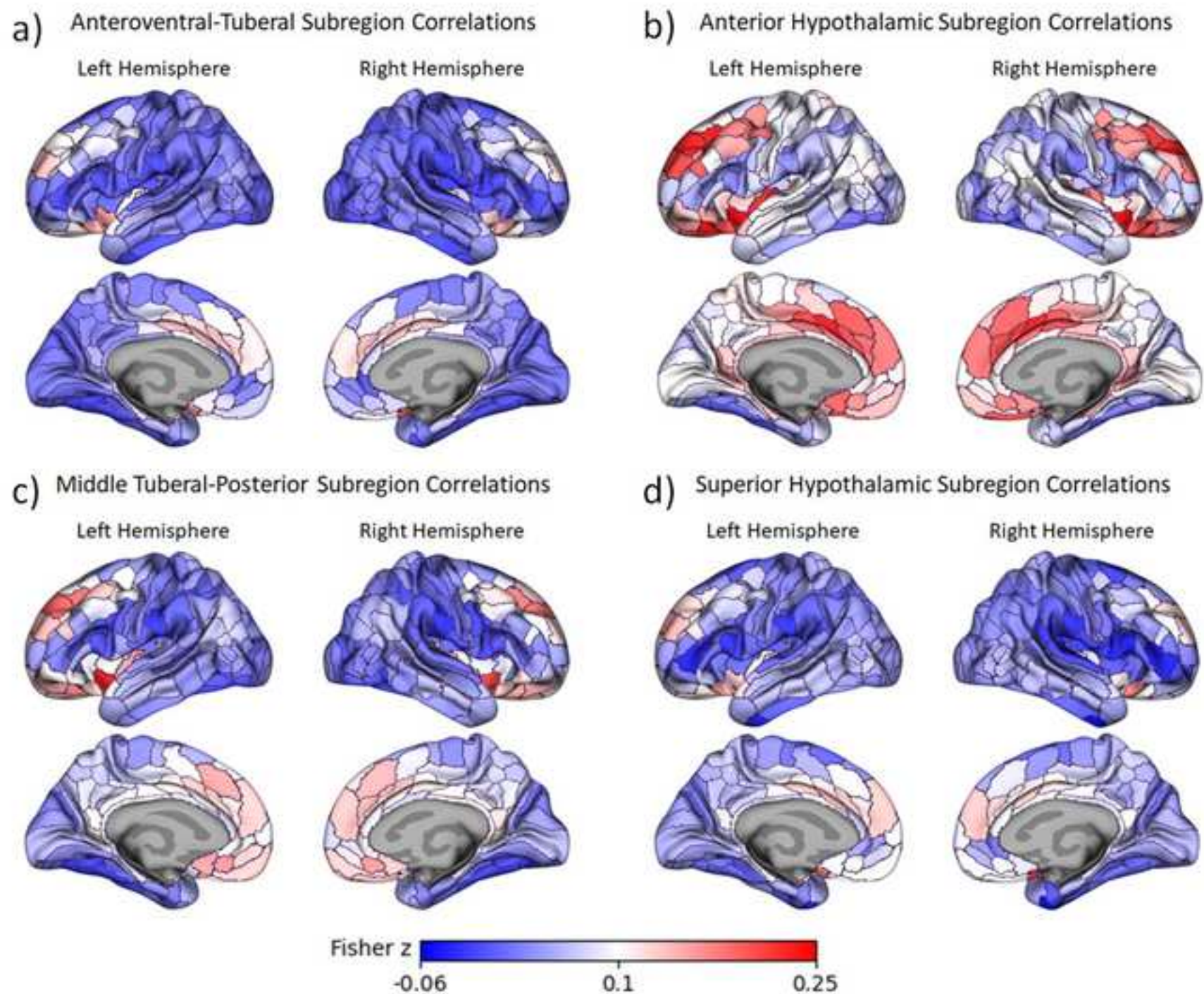


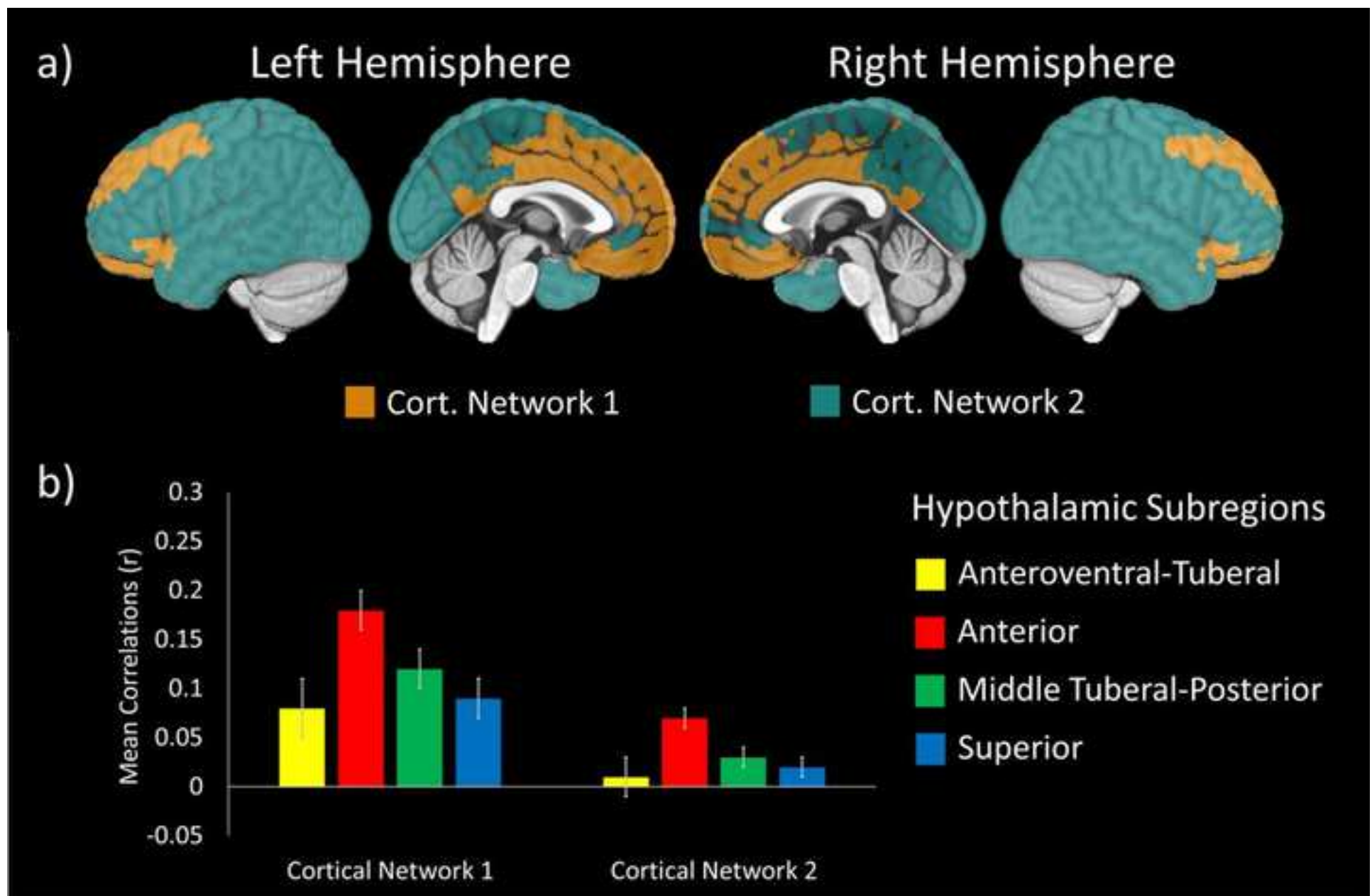
Supplementary Figure S1. Elbow plot of optimal number of clusters (Cortical Networks). The plot shows the Calinski-Harabasz index score on the Y-axis as the number of clusters (Cortical Networks) increases on the X-axis. Higher index scores reflect better clustering, as defined by greater between cluster variances vs. within cluster variance. The optimal number of clusters identified by the Kneedle algorithm was two, indicated by the red dashed line.

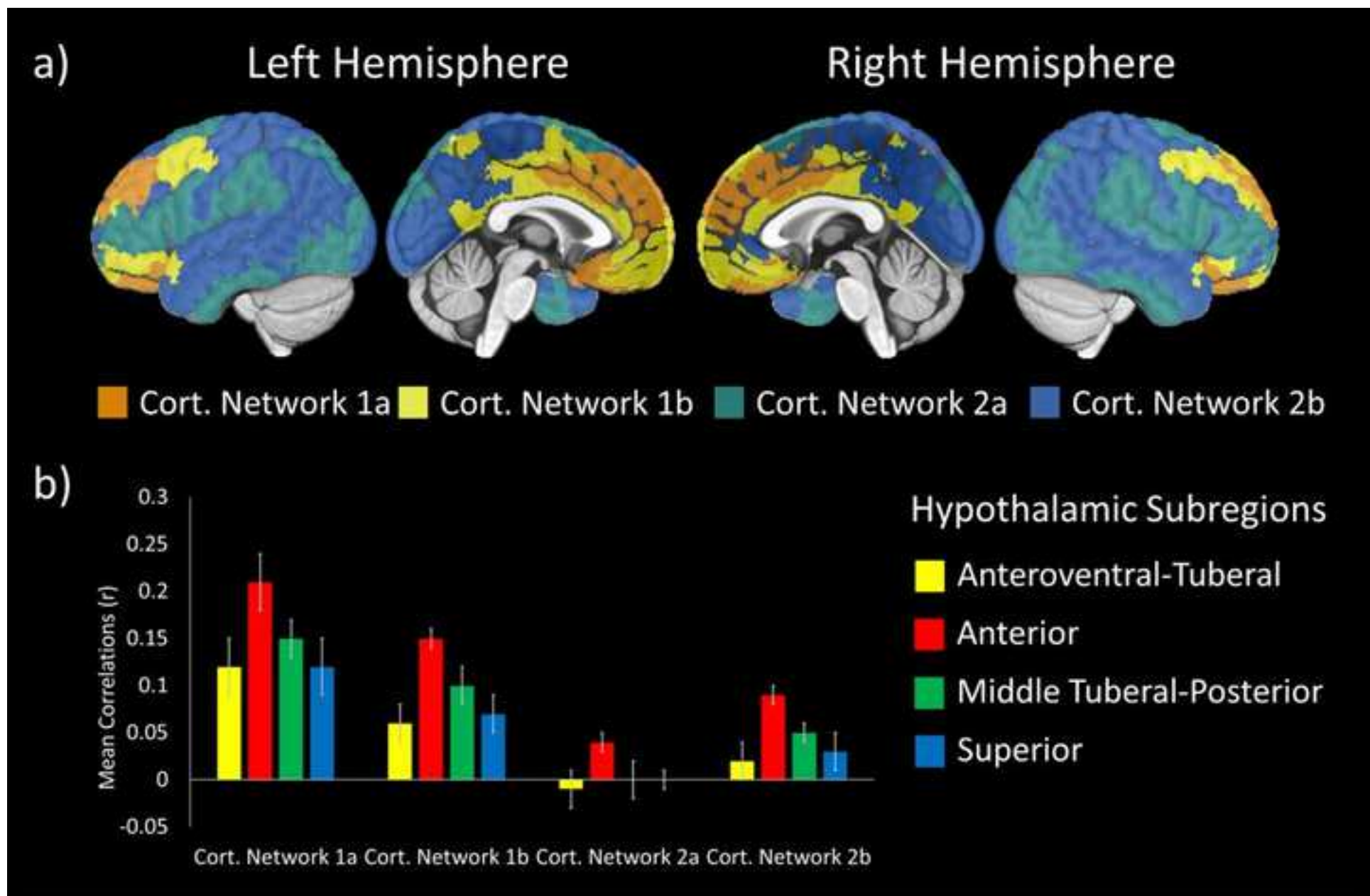


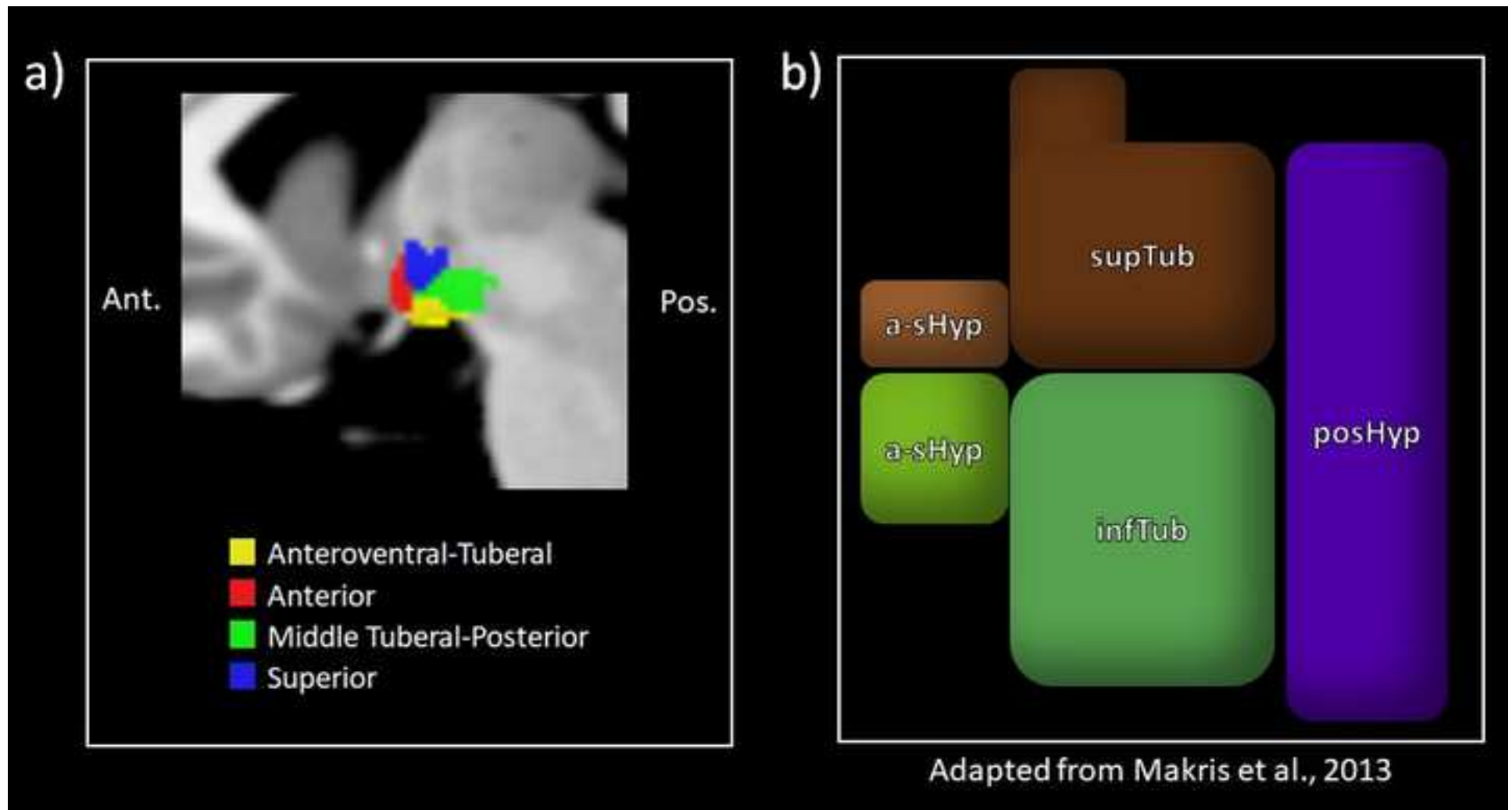
Supplementary Figure S2. Count of the number of communities/subregions identified by Louvain community detection after 1000 iterations. The Y-axis shows the frequency of the number of communities/subregions identified by Louvain community detection while the X-axis shows the number of subregions identified.











Author Summary

Despite its importance in metabolic regulation and behavior, the functional architecture of the human hypothalamus has been surprisingly understudied. Functional connectivity studies have tended to treat the hypothalamus as a unitary structure. Yet, the hypothalamus is anatomically heterogeneous which raises questions about whether functional subunits exist within it, and their relationship with the cortex. Using resting state data from ultra-highfield (7-Tesla) functional MRI, we identified four functional hypothalamic subunits. Notably, the anterior hypothalamus, compared with other subregions, showed greater connectivity with a network of frontal and midline structures, including default mode network (DMN) regions. Our work both raises questions about how the hypothalamus and DMN regions might interact and whether the hypothalamus is better thought of as functionally heterogeneous.